Supplementary Material for "Kinetic vs. Thermodynamic Control of Mutational Effects on Protein Homeostasis: A Perspective from Computational Modeling and Experiment"

Kristine Faye R. Pobre,[1] David L. Powers,[2] Kingshuk Ghosh,[3] Lila M. Gierasch,[1,*] Evan T. Powers[4,*]

[1]Departments of Chemistry and Biochemistry & Molecular Biology, University of Massachusetts-Amherst, Amherst, MA 01003, USA

[2]Department of Mathematics, Clarkson University, Potsdam, NY 13699, USA

[3]Department of Physics and Astronomy, University of Denver, Denver, CO 80208

[4]Department of Chemistry, The Scripps Research Institute, La Jolla, CA 92037, USA

Figure S1. A contour plot showing the effects of mutations on proteostasis as measured by the fraction of a test protein derived from cluster 12 that achieves the native state at steady state ($f_{N,test\text{-}12}$). The thermodynamic effects of the mutations are measured on the x-axis by the change in the folding equilibrium constant ($\Delta \ln K_{f,test\text{-}12}$; note that the corresponding change in free energy is shown along the top of the plot). The kinetic effects of the mutations are measured by the φ-values of the mutations (where $\varphi = \Delta \ln k_{f,test\text{-}12} / \Delta \ln K_{f,test\text{-}12}$). Two sets of contours are shown. The solid contours are for constant values of $f_{N,test\text{-}12}$. The dashed contours are for $\Delta \ln k_{f,test\text{-}12}$. Note that these two sets of contours almost overlay each other.

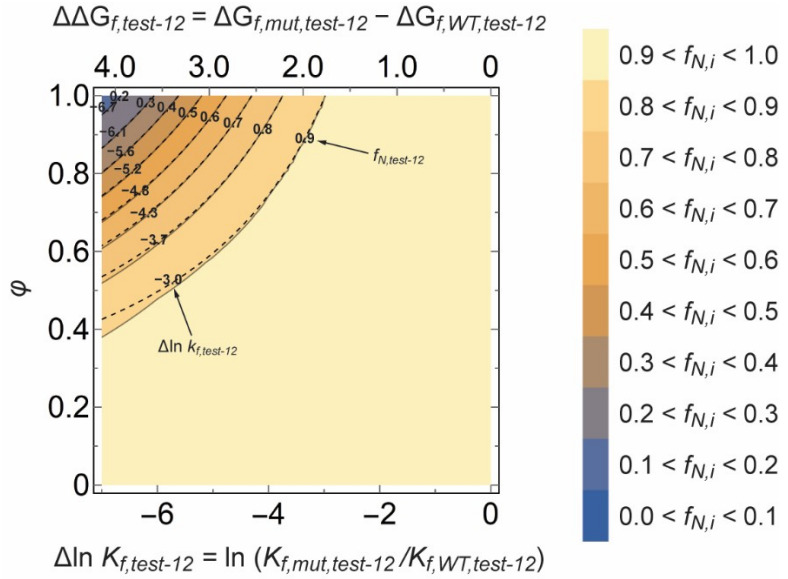


Figure S2. Time courses of expression of WT* and Y133S MmCRABP1 in *E. coli* with or without the gene for Lon protease. Red lines and data points: total MmCRABP1 concentrations per cell. Blue lines and data points: soluble MmCRABP1 concentrations per cell. The difference between the total and soluble protein concentrations is the aggregated protein concentration (red shaded area). The fractions of soluble protein at the 2 h time point for all variants are indicated on the plots. Error bars indicate the standard errors of the mean.

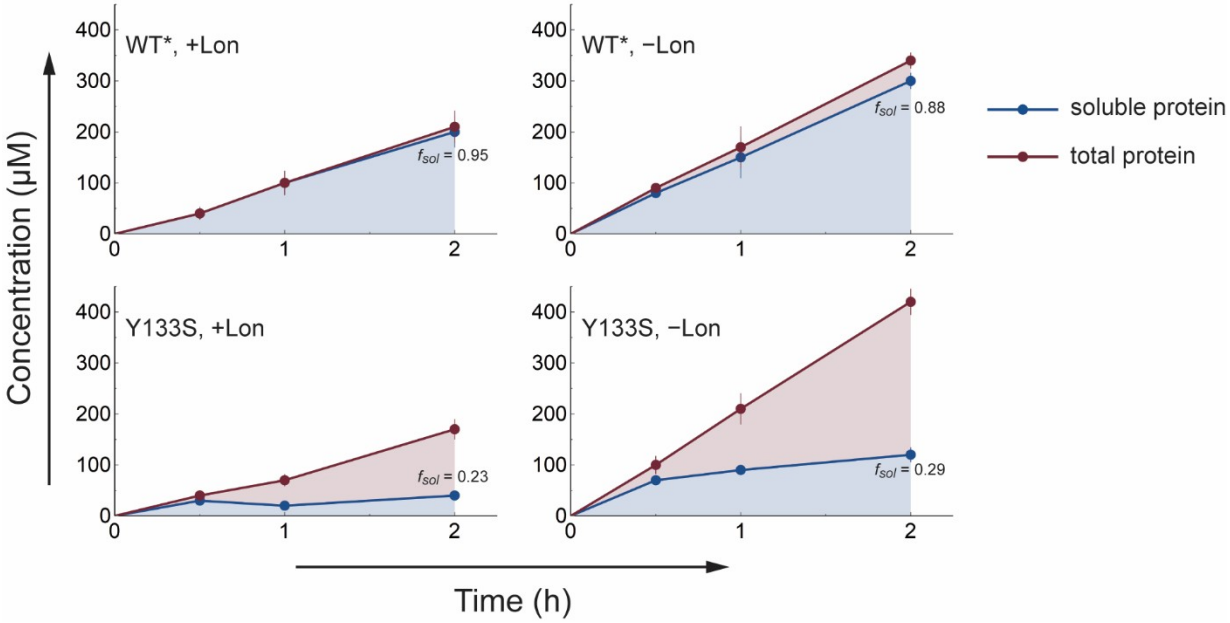


Table S1. Protein-specific folding and chaperoning parameters and fractional abundances for the clusters in our coarse-grained proteome model. See also Figure 2.

Cluster (category[a])	$k_{f,i}$ (s^{-1})	$k_{u,i}$ (s^{-1})	$K_{f,i}$	$\Delta G_{f,i}$ (kcal mol^{-1})	$k_{agg,i}$ (s^{-1})	$k_{K,disagg,i}$ (s^{-1})	$k_{G,in,i}$ (μM^{-1} s^{-1})	Abundance
1 (3)	130	1.7×10^{-5}	7.4×10^{6}	−9.4	8.5	0.011	1.0×10^{-3}	0.078
2 (1)	430	0.29	1500	−4.3	0.044	0.014	9.0×10^{-4}	0.040
3 (1)	87	0.016	5600	−5.1	0.020	0.011	1.0×10^{-3}	0.133
4 (1)	440	0.57	780	−3.9	8.5×10^{-6}	0.011	1.0×10^{-3}	0.059
5 (1)	14	0.0030	4700	−5.0	0.050	0.011	1.0×10^{-3}	0.081
6 (1)	18	0.074	250	−3.3	0.0025	0.011	1.0×10^{-3}	0.047
7 (4)	35	0.014	2500	−4.6	0.82	0.017	12	0.031
8 (2)	0.83	3.0×10^{-9}	2.8×10^{8}	−11.5	0.055	0.012	1.1×10^{-3}	0.207
9 (3)	2.5	7.0×10^{-6}	3.5×10^{5}	−7.6	1.3	0.065	3.7	0.032
10 (3)	0.05	3.7×10^{-6}	1.3×10^{4}	−5.6	0.0029	0.011	1.0×10^{-3}	0.035
11 (1)	0.30	0.0047	64	−2.5	0.0013	0.011	1.0×10^{-3}	0.073
12 (2)	1.0	2.1×10^{-6}	4.9×10^{5}	−7.8	1.3×10^{-5}	0.011	1.0×10^{-3}	0.103
13 (1)	2.0	0.0052	380	−3.5	0.0076	0.011	1.0×10^{-3}	0.009
14 (1)	13	0.00067	2.0×10^{4}	−5.8	0.26	0.011	1.0×10^{-3}	0.029
15 (1)	3700	8.9	410	−3.6	0.0031	0.011	1.0×10^{-3}	0.029
16 (2)	0.07	1.5×10^{-7}	4.9×10^{5}	−7.8	0.36	0.0019	60	0.014

[a]Mutation response category as defined by the how $f_{N,i}$ depends on the thermodynamic ($\Delta\ln K_{f,i}$) and kinetic (φ) effects of a mutation as manifested in the shapes of the contour lines in Figure 3.

Table S2. Protein-dependent folding and chaperoning parameters for the proteome. This table is contained in the associated Excel file.

Supplementary Methods

Rate Equations for FoldEcoSlim

The rate equations for FoldEcoSlim for each protein or cluster of proteins "i" are as follows:

$$d[U_i]/dt = \sigma_i - [U_i](k_{f,i} + k_{agg,i} + k_{K,in}[K] + k_{G,in,i}[G] + k_{D,in}[D] + \lambda) + k_{U,i}[N_i] + k_{K,out}[K:U_i] + k_{G,out}[G:U_i] + k_{D,out}[D:U_i]$$ 11* MERGEFORMAT (S)

$$d[N_i]/dt = k_{f,i}[U_i] - k_{a,i}[N_i] + k_{G,out}[G:N_i] - \lambda[N_i]$$ 22* MERGEFORMAT (S)

$$d[A_i]/dt = k_{agg,i}[U_i] - k_{K,in}[K][A_i] + k_{K,out}[K:A_i] - \lambda[A_i]$$ 33* MERGEFORMAT (S)

$$d[Deg_i]/dt = k_{D,deg}[D:U_i] - \lambda[Deg_i]$$ 44* MERGEFORMAT (S)

$$d[D:U_i]/dt = k_{D,in}[D][U_i] - k_{D,out}[D:U_i] - k_{D,deg}[D:U_i] - \lambda[D:U_i]$$ 55* MERGEFORMAT (S)

$$d[G:U_i]/dt = k_{G,in,i}[G][U_i] - k_{G,out}[G:U_i] - \gamma_i k_{f,i}[G:U_i] + k_{a,i}[G:N_i] - \lambda[G:U_i]$$ 66* MERGEFORMAT (S)

$$d[G:N_i]/dt = \gamma_i k_{f,i}[G:U_i] - k_{a,i}[G:N_i] - k_{G,out}[G:N_i] - \lambda[G:N_i]$$ 77* MERGEFORMAT (S)

$$d[K:U_i]/dt = k_{K,in}[K][U_i] - k_{K,out}[K:U_i] + k_{K,disagg,i}[K:A_i] - \lambda[K:U_i]$$ 88* MERGEFORMAT (S)

$$d[K:A_i]/dt = k_{K,in}[K][A_i] - k_{K,out}[K:A_i] - k_{K,disagg,i}[K:A_i] - \lambda[K:A_i]$$ 99* MERGEFORMAT (S)

All rate constants are defined in Figure 1. [Deg] is the concentration of protein per cell that has been degraded. The rate equations for the proteostasis network components H, D, G, and K are as follows:

$$d[H]/dt = \sigma_H - k_{K,in,H}[K][H] + k_{K,out,H}[K:H] - \lambda[H]$$ 1010* MERGEFORMAT (S)

$$d[K:H]/dt = k_{K,in,H}[K][H] - k_{K,out,H}[K:H] - k_{K:H,deg}[K:H] - \lambda[K:H]$$ 1111* MERGEFORMAT (S)

$$d[D]/dt = \sigma_D - \sum_{i=\text{all proteins}} k_{D,in}[D][U_i] + \left(k_{D,out} + k_{D,deg}\right) \sum_{i=\text{all proteins}} [D:U_i] - \lambda[D]$$

1212*

MERGEFORMAT (S)

$$d[G]/dt = \sigma_G - \sum_{i=\text{all proteins}} k_{G,in,i}[G][U_i] + \sum_{i=\text{all proteins}} k_{G,out}\left([G:U_i] + [G:N_i]\right) - \lambda[G]$$

1313*

MERGEFORMAT (S)

$$d[K]/dt = \sigma_K - k_{K,in,H}[K][H] + k_{K,out,H}[K:H] + k_{K:H,deg}[K:H] - \sum_{i=\text{all proteins}} k_{K,in}[K]\left([U_i] + [A_i]\right) + \sum_{i=\text{all proteins}} k_{K,out}\left([K:U_i] + [K:A_i]\right) - \lambda[K]$$

1414*

MERGEFORMAT (S)

Again, the rate constants in these equations are defined in Figure 1. The synthesis rates for D, G, and K, symbolized above by σ_D, σ_G, and σ_K, are each proportional to the concentration of the transcription factor H; thus, $\sigma_D = \rho_D[H]$, $\sigma_G = \rho_G[H]$, and $\sigma_K = \rho_K[H]$, where the quantities ρ represent the responsiveness of the promoters for D, G, and K to transcription mediated by H. The values of σ_H, $k_{K:H,deg}$, ρ_D, ρ_G, and ρ_K are set to yield steady state concentrations of D, G, and K that roughly match the measured concentrations of Lon, GroEL, and DnaK respectively in the context of the background proteome. The values used for protein-dependent parameters (i.e., those that have a subscript "i" in the rate equations above) are given in Table S1. The values used for protein-independent parameters are as follows: $\lambda = 0.00053\ s^{-1}$; $\sigma_H = 0.00318\ \mu M\ s^{-1}$; $k_{K,in,H} = 1\ \mu M^{-1}\ s^{-1}$; $k_{K,out,H} = 0.1\ s^{-1}$; $k_{K:H,deg} = 0.01\ s^{-1}$; $\rho_D = 0.000533\ s^{-1}$; $\rho_G = 0.002667\ s^{-1}$; $\rho_K = 0.0333\ s^{-1}$; $k_{D,in} = 0.1\ \mu M^{-1}\ s^{-1}$; $k_{D,out} = 0.1\ s^{-1}$; $k_{D,deg} = 0.2\ s^{-1}$; $k_{G,out} = 0.1\ s^{-1}$; $k_{K,in} = 1\ \mu M^{-1}\ s^{-1}$; $k_{K,out} = 0.1\ s^{-1}$. These parameters values were chosen to reflect the values used in our previous work with FoldEco.[1] For each protein or protein cluster i, the synthesis rate is equal to the protein's fractional proteomic abundance times the overall (non-proteostasis network) protein synthesis rate, or $\sigma_i = f_{abundance,i} \times \sigma_{tot}$ where $\sigma_{tot} = 3.975\ \mu M\ s^{-1}$.

Parametrizing the Background Proteome

Modeling the background proteome requires estimates of key parameters for as many proteins as possible, including abundances, folding and aggregation rate constants, and

competence for chaperoning. New mass spectrometry-based methods have enabled such proteome-wide estimates to be made for protein stabilities in a number of systems, including *E. coli*.[2,3] These measurements of folding equilibrium constants ($K_{f,i}$) can be combined with estimates of protein folding rate constants ($k_{f,i}$) to calculate unfolding rate constants ($k_{u,i}$; since $k_{u,i} = k_{f,i}/K_{f,i}$). The necessary folding rate constants are not available from direct experimental measurements, but it has been shown that the topologies of protein structures explain much of the variability in folding rate constants, so this information can be used to predict folding rate constants for proteins whose structure is known.[4-8] Thus, folding rate constants were estimated as described previously[8] and used in combination with folding equilibrium constants to calculate unfolding rate constants for the hundreds of *E. coli* proteins for which X-ray crystal structures have been reported. For multi-domain proteins, the rate constant for the slowest folding domain was used, since this would likely be the rate limiting step in the overall folding of such a protein.

With estimates of $k_{f,i}$ and $k_{u,i}$ in hand, proteome-wide solubility measurements made in cell-free translation systems by Niwa and co-workers[9] can be used to estimate rate constants for aggregation, $k_{agg,i}$. Since these solubility measurements were made in the absence of chaperones and proteases, FoldEcoSlim can be used with all parameters set to 0 except $k_{f,i}$ and $k_{u,i}$ and the synthesis rate (σ_i, which can be determined from the total amount of protein synthesized in the experiment) and solved for the value of $k_{agg,i}$ that reproduces the observed extent of aggregation. A complete set of estimates for $k_{f,i}$, $k_{u,i}$, and $k_{agg,i}$ were obtained in this way for 640 proteins representing 37% of the *E. coli* proteome by mass.

The aptitude for chaperoning was measured for many of the less soluble proteins examined by Niwa and co-workers by adding the DnaK/DnaJ/GrpE or GroEL/GroES systems to the same cell-free translation system and measuring the increase in solubility.[10] Given the wide distribution of DnaK binding sites in the *E. coli* proteome (on average, each protein has over 20),[11] we assumed that the rate constant for the initial binding step ($k_{K,in}$) is substrate independent, as we[1,12] and others[13,14] have done previously. Thus, we account for a protein's aptitude for chaperoning by K entirely with the parameter $k_{K,disagg,i}$, the rate constant for conversion of K:A_i to K:U_i. Using the estimates of $k_{f,i}$, $k_{u,i}$, and $k_{agg,i}$ obtained as described above,

$k_{K,disagg,i}$ was varied to find the value that reproduced the observed increase in solubility in the presence of the DnaK/DnaJ/GrpE system. In contrast to the DnaK/DnaJ/GrpE system, the GroEL/GroES system seems to be more selective in its substrates. This assertion is consistent with the observation that only ~250 proteins co-immunoprecipitate with GroEL[15] (of which ~60 are obligate substrates[16]). Thus, we varied the rate constant for substrates entering the GroEL/GroES cycle ($k_{G,in,i}$) to find the value that reproduced the observed increase in solubility in the presence of the GroEL/GroES system. There were 120 proteins for which estimates of $k_{f,i}$, $k_{u,i}$, and $k_{agg,i}$ as well as measurements of chaperone efficacy were available. The remaining proteins were well behaved in the cell free translation system,[9,10] and therefore unlikely to be substantial contributors to the burden on the proteostasis network. For modeling purposes, they were assigned the median value of the estimates for $k_{K,disagg,i}$ (0.0037 s^{-1}) and a minimal value for $k_{G,in,i}$ (10^{-3} M^{-1} s^{-1}).

Individually tracking each of the 640 proteins for which full parameter sets were available would be excessively computationally expensive, even in a relatively simple model like FoldEcoSlim. We therefore coarse-grained the proteome by clustering the proteins into 16 groups based on the similarities between their parameters. The clusters were assigned the abundance-weighted mean parameters of their members. The fractional abundance of the clusters was calculated by dividing the sum of the abundances of the cluster members (as determined by mass spectrometry by Schmidt and co-workers for *E. coli* growing in LB media[17]) by the sum of the abundances of all the proteins in the dataset. One cluster (cluster 16) was set aside to contain the obligate GroEL substrates, as defined by Fujiwara and co-workers.[16]

The resulting proteome produced an unrealistically high amount of protein aggregation (~12% of the proteome). However, there are chaperone systems in *E. coli* that suppress aggregation *in vivo* that are not included in FoldEcoSlim (for example, ClpB[18] and IbpA/IbpB[19]). In addition, it is possible that the cytosolic environment is more solubilizing for proteins than the environment of the *in vitro* translation system in which the solubilities were originally measured. To account for these factors, we found that a decrease in the aggregation rate constants (by a factor of about 7) and an increase in the efficiencies of the DnaK/DnaJ/GrpE cycle (by a factor of about 3) across the proteome mitigated the excessive aggregation that we

observed. The final protein-dependent parameters for the protein clusters are displayed in Figure 2 and are listed in Table S1. The un-adjusted parameters for individual proteins are listed in Table S2.

The Contribution of the GroEL/GroES System to Proteostasis Depends on the Relative Rates of Substrate Folding and Release

In terms of the pseudo-state sum, the GroEL/GroES system influences proteostasis primarily through $Q_{N,i}$ and the DnaK/DnaJ/GrpE system primarily through $Q_{A,i}$ (both systems contribute to $Q_{B,i}$, but this is not a major contributor to $Z_{SS,i}$ under normal conditions). The assistance that the GroEL/GroES system provides to protein folding is expressed in the term Γ_i in the expression for $Q_{N,i}$

$$\Gamma_i = \frac{k_{G,out}}{k_{u,i} + \lambda} \times \frac{\gamma_i k_{f,i}}{\gamma_i k_{f,i} + k_{u,i} + k_{G,out} + \lambda} \times \frac{k_{G,in,i}[G]}{k_{G,out} + \lambda}$$

1515* MERGEFORMAT (S)

For simplicity, we will assume in the analysis that follows that folding for proteins of interest is not enhanced by the GroEL cavity ($\gamma_i = 1$), and we will ignore the factor γ_i hereafter. The magnitude of Γ_i depends on the product of three factors, the first of which is $k_{G,out} / (k_{u,i} + \lambda)$. This factor represents the relative rate of release of natively folded protein from the GroEL cavity compared to the rate of unfolding of native protein. The native state can only accumulate substantially from GroEL-assisted folding if $k_{G,out} > (k_{u,i} + \lambda)$; if this relation is not true, the protein will unfold faster than it is released from GroEL.

The second factor is $k_{f,i} / (k_{f,i} + k_{u,i} + k_{G,out} + \lambda)$. This factor is equal to the fraction of the GroEL-bound protein that is in the native state and it approaches its maximum of 1 when the intra-cavity folding rate constant ($k_{f,i}$) is much faster than unfolding ($k_{u,i}$), release from the cavity ($k_{G,out}$) and cell division (λ). It decreases from 1 if either $k_{u,i}$ or $k_{G,out}$ approaches or exceeds $k_{f,i}$. The former condition would indicate native state instability. The latter, more likely condition

would indicate that folding cannot reach completion on the time scale of the ATP-driven GroEL/GroES conformational cycle.

The last factor is $k_{G,in,i}$ [G] / ($k_{G,out}$ + λ). In the numerator, the free concentration of G is largely determined by the folding load exerted by the proteome as a whole and is not strongly influenced by the properties of any individual substrate (as long as that substrate's expression level is not too high). Thus, in the analysis of the effects of mutations on a low-to-moderately expressed individual protein, [G] will be roughly constant and $k_{G,in,i}$ [G] can be treated as a pseudo-first order rate constant which we will denote $k'_{G,in,i}$. In the denominator, $k_{G,out}$ is set by the ATP hydrolysis rate of the GroEL/GroES complex (~0.1 s^{-1}),[20] which is two hundred-fold greater than λ even for very rapidly dividing cells, so $k_{G,out}$ + λ ≈ $k_{G,out}$. Given these approximations, the third factor becomes $k'_{G,in,i}$ / $k_{G,out}$, which is essentially the substrate binding equilibrium constant for the GroEL/GroES system.

Two limiting situations are of interest for GroEL chaperoning. The first is for fast-folding proteins for which $k_{f,i}$ >> $k_{u,i}$ + $k_{G,out}$ + λ and the second factor in Γ_i is ~1. In this case,

$$\Gamma_i \simeq \frac{k_{G,out}}{k_{u,i}+\lambda} \times \frac{k'_{G,in,i}}{k_{G,out}} = \frac{k'_{G,in,i}}{k_{u,i}+\lambda}$$

1616* MERGEFORMAT (S)

and $Q_{N,i}$ can be written

$$Q_{N,i} = \frac{k_{f,i}}{k_{u,i}+\lambda} + \Gamma_i \simeq \frac{k_{f,i}}{k_{u,i}+\lambda} + \frac{k'_{G,in,i}}{k_{u,i}+\lambda} = \frac{k_{f,i}}{k_{u,i}+\lambda}\left(1+\frac{k'_{G,in,i}}{k_{f,i}}\right)$$

1717* MERGEFORMAT (S)

The factor in parenthesis, 1 + $k'_{G,in,i}/k_{f,i}$ represents the increase in $Q_{N,i}$ due to the GroEL/GroES system. This result indicates that the GroEL/GroES system will be ineffective for fast-folding proteins if the pseudo-first order rate constant for substrate binding is not greater than or at least comparable to the rate constant for folding. In fact, the $k_{f,i}$ values for which this requirement is true and that meet the criterion for the approximation are quite constrained. Recall that the approximation requires that $k_{f,i}$ >> 0.1 s^{-1}, since $k_{G,out}$ = 0.1 s^{-1} and $k_{f,i}$ must be >> $k_{G,out}$ + $k_{u,i}$ + λ. Thus, for 1 + $k'_{G,in,i}/k_{f,i}$ to be substantially greater than 1, $k'_{G,in,i}$ ≥ $k_{f,i}$ >> 0.1 s^{-1}. This requires very fast binding to GroEL, especially considering that the steady state concentration

of G is ~95 nM, so $k_{G,in,i} \geq k_{f,i}/[G] >> 0.1\ s^{-1} / 95\ nM = 1.05 \times 10^6\ M^{-1}\ s^{-1}$. Thus, the bimolecular rate constant for substrate binding would have to be at least $10^7\ M^{-1}\ s^{-1}$ for a fast-folding protein to benefit significantly from GroEL chaperoning. This circumstance is not implausible, but none of the clusters in our model of the proteome meet all these criteria. In other words, there are no fast-folding proteins in our model that benefit greatly from the GroEL/GroES system. The closest are the proteins in cluster 7, for which $k_{f,i} = 35\ s^{-1}$ (so the approximation $k_{f,i} >> k_{u,i} + k_{G,out} + \lambda$ is valid), but $k'_{G,in,i} = k_{G,in,i}[G] = 1.2 \times 10^7\ M^{-1}\ s^{-1} \times 95\ nM = 1.1\ s^{-1}$, so $1 + k'_{G,in}/k_f$ is only 1.031. Substrate binding is not fast enough in this cluster to contribute substantially to unassisted folding and the GroEL system only increases $Q_{N,i}$ by ~3%. We note, however, that this situation can change for mutants of this protein, which can explain the unique appearance of the contour plots of $f_{N,i}$ as a function of $\Delta \ln K_{f,i}$ and φ for cluster 7 in Figure 3; see below for further discussion.

The second limiting situation is for proteins that fold slowly on the time scale of substrate release from the GroEL cavity ($k_{G,out} >> k_{f,i} + k_{u,i} + \lambda$) and the second factor in Γ_i is therefore $\sim k_{f,i}/k_{G,out}$. Recalling that we have assumed that $\gamma_i = 1$, $k_{G,in,i}[G] \sim k'_{G,in,i}$, and $k_{G,out} + \lambda \sim k_{G,out}$, this limiting case leads to the following approximation for Γ_i:

$$\Gamma \simeq \frac{k_{G,out}}{k_{u,i} + \lambda} \times \frac{k_{f,i}}{k_{G,out}} \times \frac{k'_{G,in,i}}{k_{G,out}} = \frac{k_{f,i}}{k_{u,i} + \lambda} \times \frac{k'_{G,in,i}}{k_{G,out}}$$

1818* MERGEFORMAT (S)

and for $Q_{N,i}$:

$$Q_{N,i} = \frac{k_{f,i}}{k_{u,i} + \lambda} + \Gamma_i \simeq \frac{k_{f,i}}{k_{u,i} + \lambda} + \frac{k_{f,i}}{k_{u,i} + \lambda} \times \frac{k'_{G,in,i}}{k_{G,out}} = \frac{k_{f,i}}{k_{u,i} + \lambda}\left(1 + \frac{k'_{G,in,i}}{k_{G,out}}\right)$$

1919* MERGEFORMAT (S)

Again, the factor in parenthesis, $1 + k'_{G,in,i}/k_{G,out}$ represents the increase in $Q_{N,i}$ due to the GroEL/GroES system. This result indicates that the effect of the GroEL/GroES system is simply proportional to its ability to bind the substrate. In this regime, the GroEL/GroES system provides a parallel channel for folding the protein of interest, but in an environment in which the protein is safeguarded from aggregation and degradation. Thus, the flux from the unfolded to the native state increases by a factor proportional to the amount of unfolded protein to which the GroEL/GroES system can bind. When substrate binding by the GroEL/GroES system is

favorable, this factor can be substantial. The proteins in cluster 16—which, as noted above, contains the obligate GroEL/GroES substrates—bind substantially to the GroEL/GroES system and have $\gamma_i k_{f,i} + k_{u,i} << k_{G,out}$. For the proteins in this cluster $1 + k'_{G,in,i}/k_{G,out} = 48.3$, so $Q_{N,i}$ is magnified by a factor of almost fifty which enables 93% of the protein in this cluster to be natively folded at steady state ($f_{N,i} = 0.93$). In the absence of the GroEL/GroES system, $f_{N,i}$ would be 0.28 and more than half of the protein that is synthesized would aggregate.

Comparing equations 17 and 19, it is clear that the effect of GroEL on proteostasis is greater for proteins that fold slowly. For a protein with a given ratio of $k_{f,i}/k_{u,i}$—that is, with a given thermodynamic stability—that falls into the fast-folding regime described above ($k_{f,i} >> k_{u,i} + k_{G,out} + \lambda$) the value of $Q_{N,i}$ will increase as $k_{f,i}$ decreases (recall that $k_{u,i}$ also decreases in this scenario to maintain a constant $K_{f,i}$). At first this will be because the GroEL/GroES multiplier for the protein, the quantity $1 + k_{G,in,i}/k_{f,i}$, increases. As $k_{f,i}$ continues to decrease, however, the protein will eventually cross over from the fast- to the slow-folding regime and the multiplier will asymptotically approach $1 + k_{G,in,i}/k_{G,out}$. This behavior is observed in the plot of $f_{N,i}$ vs. $\Delta\ln K_{f,i}$ and φ for cluster 7 in Figure 3. The curvatures of the contours at higher φ-values indicate that highly destabilizing mutations are less deleterious when they primarily diminish the folding rate. Thus, the GroEL/GroES system inverts the usual effects of mutations.

The DnaK/DnaJ/GrpE System Is the Primary Means of Suppressing Aggregation Even in Fast-Dividing Bacteria

The DnaK/DnaJ/GrpE system influences proteostasis by dispersing aggregates. It can also bind to unfolded proteins, but, rather than helping them to fold, it tends to release them in a state that is still unfolded (if anything, it is an even more expanded state[21-23]). Thus, among the steady state constants in the pseudo-state sum of a given protein, the DnaK/DnaJ/GrpE system contributes to $Q_{A,i}$ (for aggregates) and $Q_{B,i}$ (for chaperone-bound proteins). As noted above, $Q_{B,i}$ rarely makes substantial contributions to $Z_{SS,i}$; thus, we will focus on $Q_{A,i}$.

The impact of aggregation on proteostasis of a protein of interest is proportional to $Q_{A,i}$; as $Q_{A,i}$ increases, the extent of aggregation increases since

$$f_{A,i} = \frac{Q_{A,i}}{Z_{ss,i}} = \frac{Q_{A,i}}{1 + Q_{N,i} + Q_{A,i} + Q_{B,i} + Q_{D,i}}$$ 2020* MERGEFORMAT (S)

where $f_{A,i}$ is the fraction of a given protein that is aggregated at steady state. Also, since $Q_{A,i}$ appears in the denominator of the expression for $f_{N,i}$, the fraction of natively folded protein (see Equation (6) in the main text). Therefore, $f_{N,i}$ decreases as $Q_{A,i}$ increases. As shown in Equation (3) in the main text, $Q_{A,i}$ increases as the rate constant for aggregation, $k_{agg,i}$, increases. This is balanced by two processes that can diminish aggregation: dilution by cell growth, represented by the rate constant λ in the denominator of $Q_{A,i}$; and disaggregation by the DnaK/DnaJ/GrpE system, represented by the effective rate constant κ_i, where

$$\kappa_i = \frac{k_{K,in}[K] \times k_{K,disagg,i}}{k_{K,in}[K] + k_{K,out} + k_{K,disagg,i} + \lambda}$$ 2121* MERGEFORMAT (S)

Since the denominator of $Q_{A,i}$ is simply the sum of κ_i and λ (see Equation (3)) the relative contributions of these two processes to mitigating protein aggregation manifest themselves in the relative values of the parameters that represent them.

We have been using λ = 0.00053 s^{-1} for bacteria under optimal conditions for cell growth, corresponding to a doubling time of ~20 minutes. Most of the parameters in κ_i are protein-independent constants, like $k_{K,in}$, $k_{K,out}$ and λ. In addition, the free concentration of K, like that of G, is largely determined by the proteomic burden of unfolded protein and is unlikely to be affected by a mutation in a protein that is expressed at a low to moderate level. Thus, $k_{K,in}$[K] can be treated as a pseudo-first order rate constant when analyzing the behavior of mutant test proteins, where $k_{K,in}$ [K] = 1 μM^{-1} s^{-1} × 64 nM = 0.064 s^{-1}. Also, $k_{K,out}$ = 0.1 s^{-1} and as noted above λ = 0.00053 s^{-1}. With this information, κ_i can be expressed as follows

$$\kappa_i = \frac{0.064\ s^{-1} \times k_{K,disagg,i}}{0.164\ s^{-1} + k_{K,disagg,i}}$$ 2222* MERGEFORMAT (S)

The parameter $k_{K,disagg,i}$, representing the efficiency with which the DnaK/DnaJ/GrpE system disperses aggregates, is protein dependent. However, for 14 out of the 16 clusters in our proteome model, $k_{K,disagg,i}$ is between 0.011 s^{-1} and 0.017 s^{-1} and κ_i is therefore between 0.004 s^{-1}

and 0.006 s^{-1}. Importantly, this is about 8- to 10-fold larger than λ, so λ can be ignored without too great a loss in accuracy in the denominator of $Q_{A,i}$. In other words, $Q_{A,i} \approx k_{agg,i}/\kappa$. This analysis shows that the extent of aggregation is primarily determined by the balance between the rates of aggregation and disaggregation, and that dilution of aggregates by cell division is a much less efficient mechanism for suppressing aggregation than the DnaK/DnaJ/GrpE system, even in fast-dividing bacteria.

The clusters that do not fall in the range quoted above for $k_{K,disagg,i}$ are cluster 9, for which $k_{K,disagg,9}$ = 0.065 s^{-1} and cluster 16, for which $k_{K,disagg,16}$ = 0.0019 s^{-1}. The former cluster therefore contains proteins that are preferred clients for the DnaK/DnaJ/GrpE system. In fact, this is the most highly aggregation-prone cluster of proteins, with aggregation being almost as fast as folding ($k_{f,9}$ = 2.45 s^{-1} vs. $k_{agg,9}$ = 1.27 s^{-1}), so disaggregation by the DnaK/DnaJ/GrpE system is particularly important for these proteins. The latter cluster represents proteins that are very inefficiently disaggregated by the DnaK/DnaJ/GrpE system, but these proteins are, probably not coincidentally, the favored substrates of the GroEL/GroES system.

Protein Degradation Can Suppress Aggregation, but Suppresses Native Folding Equally

Proteases like Lon in *E. coli* dispose of proteins that are not folded and are therefore vulnerable to misfolding and aggregation. The extent of degradation is determined by the steady state constant $Q_{D,i}$, which is defined in Equation (5) in the main text. In FoldEcoSlim, the value of $Q_{D,i}$ is largely protein independent. The free concentration of D, which is in the numerator of the expression for $Q_{D,i}$, should be unaffected by mutations in proteins that are not abundant. Thus, we can set [D] = 73 nM. Using $k_{D,in}$ = 0.1 s^{-1}, $k_{D,out}$ = 0.1 s^{-1}, and $k_{D,deg}$ = 0.2 s^{-1}, the value of $Q_{D,i}$ for all proteins should be

$$Q_{D,i} = \frac{k_{D,in}[D]}{\lambda}\left(\frac{k_{D,deg}}{k_{D,out}+k_{D,deg}+\lambda}\right) = \frac{0.1\ \mu M^{-1}s^{-1} \times 0.073\ \mu M}{0.00053\ s^{-1}}\left(\frac{0.2\ s^{-1}}{0.30053\ s^{-1}}\right) = 9.17$$

2323* MERGEFORMAT (S)

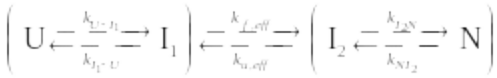
(Recall that, as discussed above, the parameters for degradation were adjusted to match the observed rate of protein turnover in *E. coli*.[24,25]) Given this value of $Q_{D,i}$, a substantial fraction of any protein with a value of $Q_{N,i}$ that is not much greater than 9.17 will be degraded. Because of this, mutations that decrease $Q_{N,i}$ can lead to excessive degradation and loss of function. In addition, it is important to note that manipulations of the proteostasis network that increase the levels of general proteases like Lon will succeed in decreasing protein aggregation because $Q_{D,i}$ is in the denominator of $f_{A,i}$ (see Equation (3)) However, $Q_{D,i}$ is also in the denominator of $f_{N,I}$ (see Equation (2)), so increasing $Q_{D,i}$ will also decrease $f_{N,i}$. The former goal cannot be achieved by increasing proteolysis without the latter consequence. For a degradation system to be able to selectively decrease aggregation, it must directly degrade aggregates as the eukaryotic autophagy system does.[26]

Fitting the MmCRABP1 Folding Mechanism into the Folding Scheme in FoldEcoSlim.

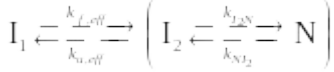
As noted in the main text, MmCRABP1 folds via three step pathway: rapid formation of an aggregation-prone hydrophobically collapsed intermediate (I_1), followed by a re-arrangement to a aggregation-resistant intermediate with native-like topology (I_2), and then a final annealing to the native state (here we ignore a low-amplitude fourth phase due to proline isomerization):[27-34]

$$\mathrm{U} \underset{k_{I_1\text{-}U}}{\overset{k_{U\text{-}I_1}}{\rightleftharpoons}} \mathrm{I}_1 \underset{k_{I_2\text{-}I_1}}{\overset{k_{I_1\text{-}I_2}}{\rightleftharpoons}} \mathrm{I}_2 \underset{k_{NI_2}}{\overset{k_{I_2N}}{\rightleftharpoons}} \mathrm{N}$$

Many of these parameters have been measured for a wide range of MmCRABP1 mutants.[27] To fit this mechanism into the folding scheme in FoldEcoSlim, we will collect the (aggregation-prone) unfolded and hydrophobic collapse states together as the U “super-state” and the (aggregation-resistant) native-like intermediate and native states together as the N super-state.



Where the $k_{f,eff}$ and $k_{u,eff}$ are the effective folding and unfolding rates between the U and N super-states. Hydrophobic collapse is rapid and greatly favors the collapsed I_1 state, so we simplify this scheme to



Thus, the effective folding rate $k_{f,eff} = k_{I1\text{-}I2}$ while $k_{u,eff}$ is equal to $k_{I2\text{-}I1}$ multiplied by the fraction of the N super-state that is actually in the I_2 form. Since the unfolding of I_2 is much slower than equilibration between I_2 and N ($k_{I1\text{-}I2} << k_{I2\text{-}N}$),[27] this fraction is $f_{I2} = 1 / (1 + K_{I2\text{-}N}) = k_{N\text{-}I2} / (k_{N\text{-}I2} + k_{I2\text{-}N})$ where $K_{I2\text{-}N} = [N]/[I_2] = k_{I2\text{-}N}/k_{N\text{-}I2}$ is the equilibrium constant between I_2 and N. Thus, $k_{u,eff} = f_{I2}\, k_{I2\text{-}I1} = k_{I2\text{-}I1}\, k_{N\text{-}I2} / (k_{N\text{-}I2} + k_{I2\text{-}N})$

All of the rate constants needed to calculate $k_{f,eff}$ and $k_{u,eff}$ for the WT*, M9A, V67A, and L118V MmCRABP1 were measured by, or can be inferred from measurements made by, Budyak and co-workers.[27] The necessary values of $k_{f,eff} = k_{I1\text{-}I2}$ are reported in Table S1 of that work (where they are denoted k_{I2}). For calculating $k_{u,eff}$, the necessary values of $k_{N\text{-}I2}$ are reported in Table 1 of that work (where they are denoted k_U) while the values of $k_{I2\text{-}N}$ are reported in Table S1 (where they are denoted k_N). For Y133S MmCRABP1, $k_{N\text{-}I2}$ is in Table 1 of Budyak and co-workers[27] and $k_{I2\text{-}N}$ is reported by Pobre[35] (see Table 2.1A of that work). The value of $k_{I1\text{-}I2}$ is not directly reported, but since the values of $k_{I1\text{-}I2}$ and $k_{I2\text{-}N}$ reported by Budyak and co-workers[27] are correlated, consistent with the transition states having similar topologies ($\ln k_{I1\text{-}I2} = 1.91 + 1.12\, k_{I2\text{-}N}$; $R^2 = 0.54$; data not shown), the value of $k_{I1\text{-}I2}$ for Y133S MmCRABP1 can be estimated from its value of $k_{I2\text{-}N}$. This estimate for $k_{I1\text{-}I2}$ for Y133S MmCRABP1 is given in Table 1 of this work.

References

(1) Powers, E. T.; Powers, D. L.; Gierasch, L. M. FoldEco: a model for proteostasis in *E. coli*. *Cell Rep*. **2012**, *1*, 265-276.

(2) Leuenberger, P.; Ganscha, S.; Kahraman, A.; Cappelletti, V.; Boersema, P. J.; von Mering, C.; Claassen, M.; Picotti, P. Cell-wide analysis of protein thermal unfolding reveals determinants of thermostability. *Science* **2017**, *355*.

(3) Mateus, A.; Bobonis, J.; Kurzawa, N.; Stein, F.; Helm, D.; Hevler, J.; Typas, A.; Savitski, M. M. Thermal proteome profiling in bacteria: probing protein state in vivo. *Mol Syst Biol* **2018**, *14*, e8242.

(4) Ivankov, D. N.; Garbuzynskiy, S. O.; Alm, E.; Plaxco, K. W.; Baker, D.; Finkelstein, A. V. Contact order revisited: influence of protein size on the folding rate. *Protein Sci* **2003**, *12*, 2057-2062.

(5) Ouyang, Z.; Liang, J. Predicting protein folding rates from geometric contact and amino acid sequence. *Protein Sci* **2008**, *17*, 1256-1263.

(6) Plaxco, K. W.; Simons, K. T.; Baker, D. Contact order, transition state placement and the refolding rates of single domain proteins. *J Mol Biol* **1998**, *277*, 985-994.

(7) Rustad, M.; Ghosh, K. Why and how does native topology dictate the folding speed of a protein? *J Chem Phys* **2012**, *137*, 205104.

(8) Zou, T.; Williams, N.; Ozkan, S. B.; Ghosh, K. Proteome folding kinetics is limited by protein halflife. *PLoS One* **2014**, *9*, e112701.

(9) Niwa, T.; Ying, B. W.; Saito, K.; Jin, W.; Takada, S.; Ueda, T.; Taguchi, H. Bimodal protein solubility distribution revealed by an aggregation analysis of the entire ensemble of Escherichia coli proteins. *Proc Natl Acad Sci U S A* **2009**, *106*, 4201-4206.

(10) Niwa, T.; Kanamori, T.; Ueda, T.; Taguchi, H. Global analysis of chaperone effects using a reconstituted cell-free translation system. *Proc Natl Acad Sci U S A* **2012**, *109*, 8937-8942.

(11) Srinivasan, S. R.; Gillies, A. T.; Chang, L.; Thompson, A. D.; Gestwicki, J. E. Molecular chaperones DnaK and DnaJ share predicted binding sites on most proteins in the E. coli proteome. *Mol Biosyst* **2012**, *8*, 2323-2333.

(12) Cho, Y.; Zhang, X.; Pobre, K. F.; Liu, Y.; Powers, D. L.; Kelly, J. W.; Gierasch, L. M.; Powers, E. T. Individual and collective contributions of chaperoning and degradation to protein homeostasis in E. coli. *Cell Rep* **2015**, *11*, 321-333.

(13) Santra, M.; Dill, K. A.; de Graff, A. M. R. How Do Chaperones Protect a Cell's Proteins from Oxidative Damage? *Cell Syst* **2018**, *6*, 743-751 e743.

(14) Santra, M.; Farrell, D. W.; Dill, K. A. Bacterial proteostasis balances energy and chaperone utilization efficiently. *Proc Natl Acad Sci U S A* **2017**, *114*, E2654-E2661.

(15) Kerner, M. J.; Naylor, D. J.; Ishihama, Y.; Maier, T.; Chang, H. C.; Stines, A. P.; Georgopoulos, C.; Frishman, D.; Hayer-Hartl, M.; Mann, M.; Hartl, F. U. Proteome-wide analysis of chaperonin-dependent protein folding in Escherichia coli. *Cell* **2005**, *122*, 209-220.

(16) Fujiwara, K.; Ishihama, Y.; Nakahigashi, K.; Soga, T.; Taguchi, H. A systematic survey of in vivo obligate chaperonin-dependent substrates. *EMBO J* **2010**, *29*, 1552-1564.

(17) Schmidt, A.; Kochanowski, K.; Vedelaar, S.; Ahrne, E.; Volkmer, B.; Callipo, L.; Knoops, K.; Bauer, M.; Aebersold, R.; Heinemann, M. The quantitative and condition-dependent Escherichia coli proteome. *Nat Biotechnol* **2016**, *34*, 104-110.

(18) Doyle, S. M.; Genest, O.; Wickner, S. Protein rescue from aggregates by powerful molecular chaperone machines. *Nat. Rev. Mol. Cell. Biol.* **2013**, *14*, 617-629.

(19) Matuszewska, M.; Kuczynska-Wisnik, D.; Laskowska, E.; Liberek, K. The small heat shock protein IbpA of Escherichia coli cooperates with IbpB in stabilization of thermally aggregated proteins in a disaggregation competent state. *J Biol Chem* **2005**, *280*, 12292-12298.

(20) Horwich, A. L.; Fenton, W. A. Chaperonin-mediated protein folding: using a central cavity to kinetically assist polypeptide chain folding. *Q. Rev. Biophys.* **2009**, *42*, 83-116.

(21) Kellner, R.; Hofmann, H.; Barducci, A.; Wunderlich, B.; Nettels, D.; Schuler, B. Single-molecule spectroscopy reveals chaperone-mediated expansion of substrate protein. *Proc Natl Acad Sci U S A* **2014**, *111*, 13355-13360.

(22) Sekhar, A.; Rosenzweig, R.; Bouvignies, G.; Kay, L. E. Mapping the conformation of a client protein through the Hsp70 functional cycle. *Proc Natl Acad Sci U S A* **2015**, *112*, 10395-10400.

(23) Sekhar, A.; Rosenzweig, R.; Bouvignies, G.; Kay, L. E. Hsp70 biases the folding pathways of client proteins. *Proc Natl Acad Sci U S A* **2016**, *113*, E2794-2801.

(24) Mandelstam, J. Turnover of protein in growing and non-growing populations of Escherichia coli. *Biochem J* **1958**, *69*, 110-119.

(25) Pine, M. J. Steady-state measurement of the turnover of amino acid in the cellular proteins of growing Escherichia coli: existence of two kinetically distinct reactions. *J Bacteriol* **1970**, *103*, 207-215.

(26) Mizushima, N.; Komatsu, M. Autophagy: renovation of cells and tissues. *Cell* **2011**, *147*, 728-741.

(27) Budyak, I. L.; Krishnan, B.; Marcelino-Cruz, A. M.; Ferrolino, M. C.; Zhuravleva, A.; Gierasch, L. M. Early folding events protect aggregation-prone regions of a beta-rich protein. *Structure* **2013**, *21*, 476-485.

(28) Clark, P. L.; Liu, Z. P.; Zhang, J.; Gierasch, L. M. Intrinsic tryptophans of CRABPI as probes of structure and folding. *Protein Sci* **1996**, *5*, 1108-1117.

(29) Clark, P. L.; Weston, B. F.; Gierasch, L. M. Probing the folding pathway of a beta-clam protein with single-tryptophan constructs. *Fold Des* **1998**, *3*, 401-412.

(30) Ignatova, Z.; Gierasch, L. M. Monitoring protein stability and aggregation in vivo by real-time fluorescent labeling. *Proc Natl Acad Sci U S A* **2004**, *101*, 523-528.

(31) Ignatova, Z.; Gierasch, L. M. Aggregation of a slow-folding mutant of a beta-clam protein proceeds through a monomeric nucleus. *Biochemistry* **2005**, *44*, 7266-7274.

(32) Kleywegt, G. J.; Bergfors, T.; Senn, H.; Le Motte, P.; Gsell, B.; Shudo, K.; Jones, T. A. Crystal structures of cellular retinoic acid binding proteins I and II in complex with all-*trans*-retinoic acid and a synthetic retinoid. *Structure* **1994**, *2*, 1241-1258.

(33) Liu, Z. P.; Rizo, J.; Gierasch, L. M. Equilibrium folding studies of cellular retinoic acid binding protein, a predominantly beta-sheet protein. *Biochemistry* **1994**, *33*, 134-142.

(34) Clark, P. L.; Liu, Z. P.; Rizo, J.; Gierasch, L. M. Cavity formation before stable hydrogen bonding in the folding of a beta-clam protein. *Nat Struct Biol* **1997**, *4*, 883-886.

(35) Pobre, K. F. R. Exploring the impact of the E. coli proteostasis network on the folding fate of proteins with different intrinsic biophysical properties. University of Massachusetts, Amherst, 2015.